RIZKY FIRMANSYAH SIPAHUTAR

+6281380482452 | rizky80firmansyah@gmail.com | Bekasi, Indonesia | LinkedIn | GitHub | Portofolio

PROFILE

A Computer Science graduate from YARSI University specializing in Artificial Intelligence, passionate about solving complex problems through technology. I have hands-on experience as an AI engineer, taking ideas from the design phase through to implementation in real-world systems using Python and SQL. With deep expertise in Deep Learning and Computer Vision, I adapt easily to new technologies, communicate technical concepts clearly, and enjoy collaborating with cross-cultural teams to design efficient, production-ready IT solutions.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher (Thesis) — Lab AI, YARSI University 08/2025 – 01/2026

CBAM-Integrated Detection Pipeline with Weighted Boxes Fusion for High-Sensitivity Cervical Cancer Screening

- Extracted and preprocessed large-scale Whole Slide Images (WSI) exceeding 1 GB from private clinical datasets, executing systematic cropping, image standardization, and cellular annotation via Roboflow.
- Engineered and optimized deep learning object detection pipelines utilizing YOLO architectures (v5, v8, v10).
- Integrated Convolutional Block Attention Module (CBAM) and Weighted Boxes Fusion (WBF) to significantly enhance feature extraction and localization of small abnormal cervical cells.
- Evaluated algorithm performance across multiple configurations using clinical statistical metrics including Precision, Recall, and mAP@50-95.
- Finalized the YOLOv5x architecture for deployment to maximize detection stability and recall on the private medical dataset.
- Collaborated closely with the Head of Anatomical Pathology to validate prediction accuracy, ensuring the AI model effectively translates technical detection into a reliable clinical decision support system.
- Deployed the optimized model into "ViuMe," an interactive digital pathology web platform, enabling real-time inference capabilities allowing end-users to visualize bounding boxes, analyze confidence scores, and perform multi-scale image analysis for clinical screening.

Machine Learning Engineer Intern — E-Health Laboratory, Universitas Yarsi Jakarta 02/2025 – 07/2025

Automating Cervical Cancer Cell Screening on Private Medical Datasets with YOLOv5x-Based Object Detection

- Extracted, cleaned, and processed large-scale raw medical image datasets to ensure high data quality and consistency for analysis.
- Conducted Exploratory Data Analysis (EDA) and data preprocessing to identify patterns and optimize features for automated screening.
- Evaluated and analyzed algorithm performance using statistical metrics (Precision, Recall, mAP) to determine the most optimal data-driven solution, finalizing on the YOLOv5x architecture.
- Collaborated with medical stakeholders to translate technical detection results and data insights into actionable medical screening solutions.

Teaching Assistant (Artificial Intelligence & OOP) — Universitas Yarsi Jakarta 04/2025 – 07/2025

- Supervised and evaluated students' final Machine Learning projects, emphasizing structured problem-solving approaches.
- Designed and delivered comprehensive lecture materials and formulated practical assignments for the Object-Oriented Programming (OOP) course.
- Mentored students during hands-on lab sessions, providing step-by-step assistance with coding exercises and assignment execution.
- Provided technical guidance on code troubleshooting and AI model performance optimization during intensive practicum sessions.
- Assisted students in the end-to-end implementation and deployment of AI models across local and server environments.

EDUCATION

Bachelor of Science in Computer Science | YARSI University
2022 - 2026

- **GPA:** *3.77 / 4.00*
- **Thesis**: *CBAM-Integrated Detection Pipeline with Weighted Boxes Fusion for High-Sensitivity Cervical Cancer Screening*
- **Relevant Courses**: *Deep Learning & Reinforcement Learning, Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems, Calculus, Probability and Statistics, Data Analytics*

AWARDS & PROJECTS

Award Cross-Cultural Collaboration | Pragma 41 Hackathon, Thammasat University
(PRAGMA Hackathon Winner)

- Successfully developed and pitched a functional AI platform for cervical cell screening during a rigorous 16-hour international hackathon.
- Collaborated effectively within a diverse cross-cultural team of undergraduate (S1) and master's (S2) students, overcoming technical barriers to deliver an award-winning prototype.

Intelligent Receipt Digitization & Accounting System (Vision-Language Models)

- Built a multimodal receipt parser using VLMs and Pydantic structured outputs, outperforming traditional OCR in extracting complex handwritten texts into strict JSON schemas.
- Automated down-payment reconciliation and debt carry-over isolation via a custom validation engine, preventing financial double-counting across reseller transactions.
- Deployed a full-stack verification interface (Next.js, TypeScript) on Vercel and Render with automated Google Sheets/Drive API data pipelines and secure GCP credential injection.

COURSES

AWS Certified AI Practitioner
Score: 719

CITI Program
Conflicts of Interest

TOIEC
Score: 540

CCNAv7: Switching, Routing, and Wireless Essentials

CCNAv7: Introduction to Networks

SKILLS

Programming & Development

- Languages: Python, SQL, JavaScript, HTML/CSS
- Backend & API Development: FastAPI, Flask REST API, RESTful APIs
- Database & Cloud Infrastructure: PostgreSQL, NoSQL, MongoDB Atlas, AWS, Docker, GPU Parallel Processing, Distributed Programming

AI & Machine Learning Engineering

- Core AI & Architectures: Deep Learning, Computer Vision (YOLO, CNN, VLM), Offline Evaluation (OPE, WIS, DR)
- ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn

Data Analytics & Visualization

- Data Processing & Analysis: Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA)
- Business Intelligence & Visualization: Power BI, Tableau, Matplotlib, Seaborn